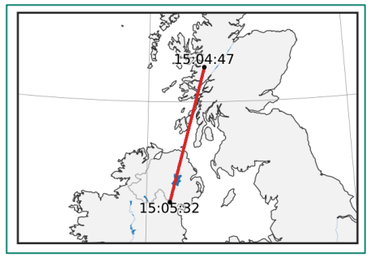
Department of Meteorology

Evaluating EarthCARE's ACM-CAP product using airborne observations

Emma Phillips | Supervisors: Chris Westbrook, Thorwald Stein, Shannon Mason

University of Reading
ECMWF

The Earth Cloud, Aerosol and Radiation Explorer (EarthCARE) is a multi-instrument satellite mission between ESA and JAXA.



EarthCARE's ACM-CAP* synergy product utilises multiple of these instruments in constraining retrievals of cloud characteristics. We aim to evaluate this product using FAAM observations, in particular, the VERIFY Campaign.

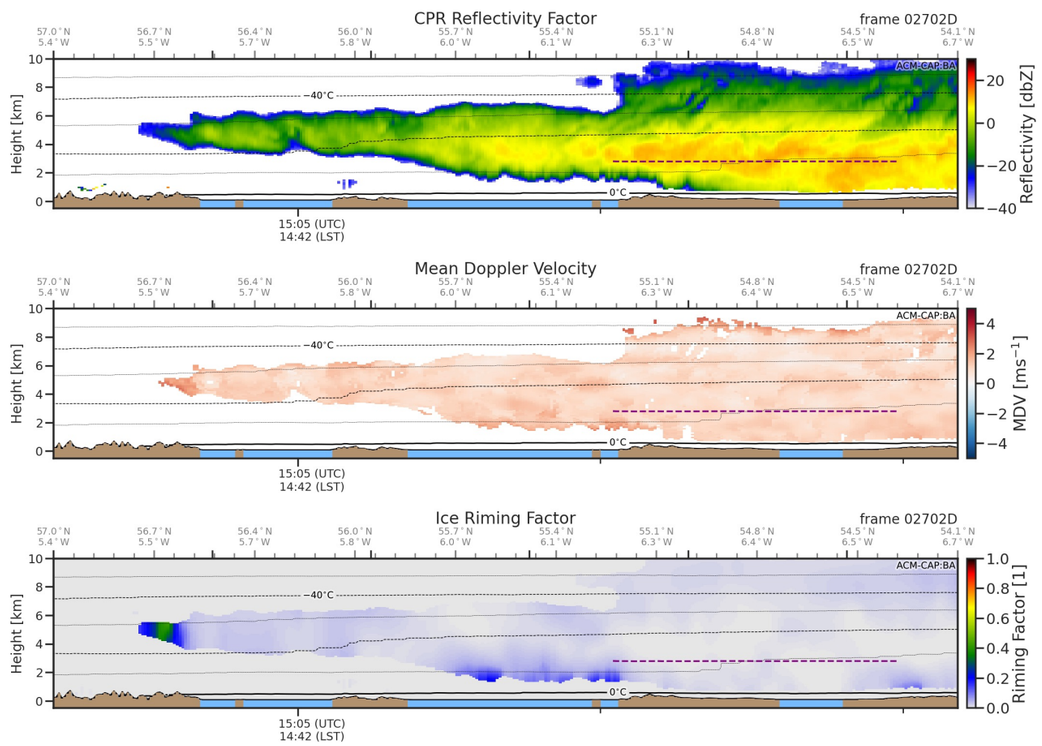


Figure 1: (top left) EarthCARE track for VERIFY flight 396.

Figure 2: EarthCARE ACM-CAP Retrieval of (a) CPR (cloud profiling radar) Reflectivity, (b) CPR Mean Doppler Velocity and (c) ACM-CAP Ice Riming Factor (*Mason, 2018*), each with VERIFY flight track for Run 5 (purple dashed line) for 18/11/2024 over the UK

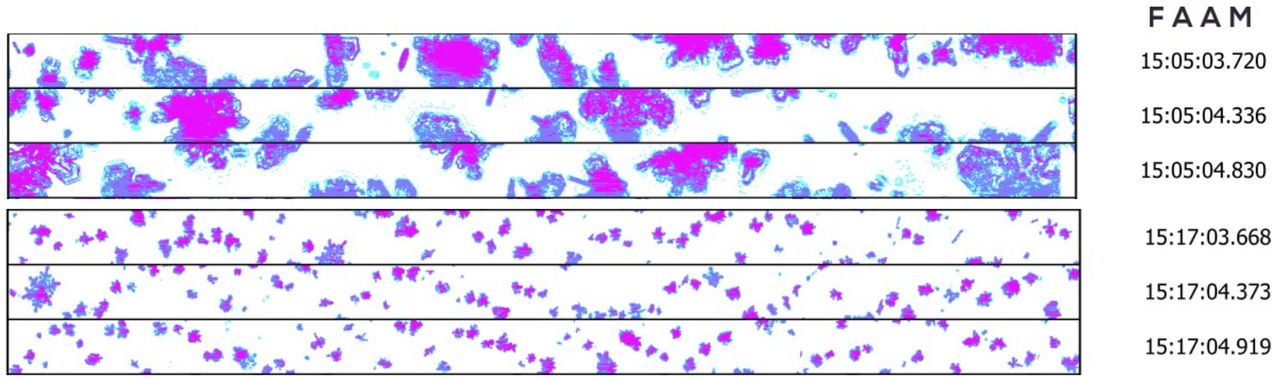


Figure 3: Ice particles imaged by CIP instrument for 15μm resolution (upper) and CIP 100μm resolution (lower) for VERIFY Flight 396 *[Altitude 2800m, Temperature -9.9ºC].*

Using the instruments onboard FAAM, including CIP (15μm and 100μm) instruments (Figure 3), we can use the measured particle spectra to model EarthCARE variables, i.e. Mean Doppler Velocity (MDV).

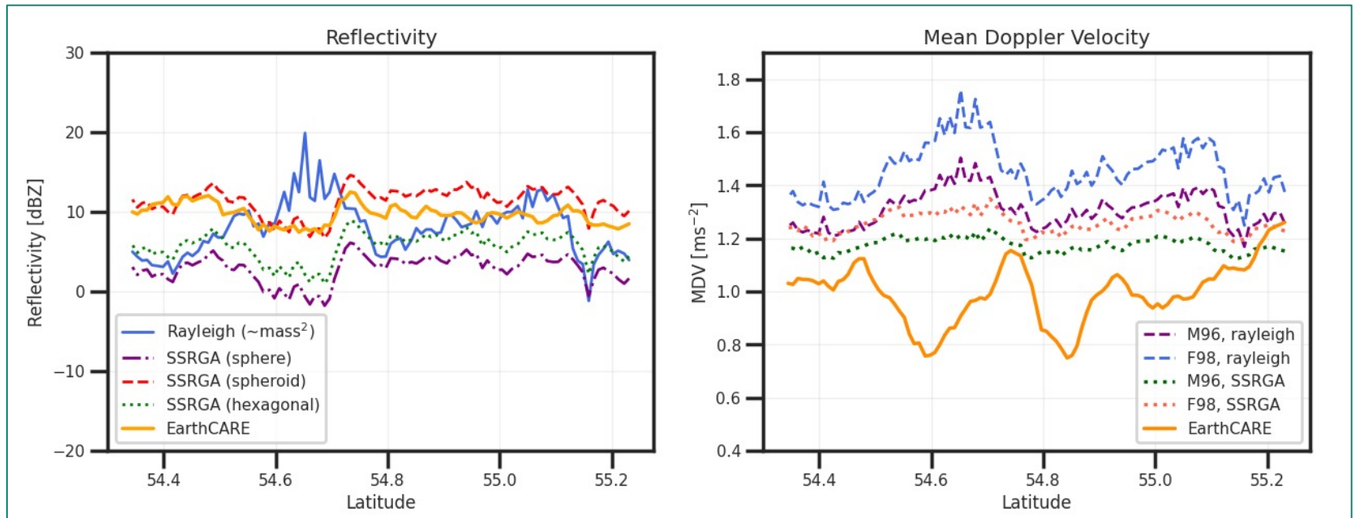


Figure 4a: Reflectivity calculated using Rayleigh Scattering approximation and SSRGA method for calculating backscatter for ice aggregates compared to CPR Reflectivity.

Figure 4b: Modelled MDV (Mean Doppler Velocity) from CIP particle size distribution compared to EarthCARE MDV from doppler radar

*ATLID-CPR-MSI Cloud-Aerosol-Precipitation

Contact: emma.phillips@pgr.reading.ac.uk